Total No. of Questions : 8]

SEAT No. :

PC2820

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S.E. (Computer Engineering)

DIGITAL ELECTRONICS AND LOGIC DESIGN

(2019 Pattern) (Semester - III) (210245)

Time : 2½ Hours] *[Max. Marks : 70*

Instructions to the candidates:

1) *Attempt Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8.*
2) *Neat diagram must be drawn wherever necessary.*
3) *Assume suitable data, if necessary.*

Q1) a) Design sequence generator using counter for sequence 1001001. **[6]**

b) Difference between sequential and combinational circuit. **[6]**

c) Explain the working of Ring counter. **[6]**

OR

Q2) a) What is MOD counter? Design MOD-6 counter using 7490. **[6]**

b) Design 2 bit synchronous up counter using MS-JK flip flop. Draw the circuit diagram. **[6]**

c) Explain in details the working of 3 bit bidirectional shift register. **[6]**

Q3) a) Draw the block diagram of PLA device and explain. **[6]**

b) Draw an ASM chart for 2 bit Up counter using mode control line **[6]**
When M = 1 : Up counting and
When M = 0 : remains in same state

c) A combinational Circuit is defined by the following function : **[5]**
$F1(A,B,C) = \Sigma\, m\, (0,1,2,4)$
$F2\,(A,B,C) = \Sigma\, m\, (1,3,5,6)$
Implement this circuit with PLA.

OR

Q4) a) Design PLA for following function : **[6]**
$F1 = A'B'C'+A'B'C+A'BC+ABC$
$F2 = A'B'C+A'BC'+AB'C+ABC'$

b) What is an ASM chart? Name the elements of an ASM chart and define each of them. **[6]**

c) Implement 4 : 1 multiplexer using suitable PAL. **[5]**

Q5) a) Compare TTL and CMOS families and also draw CMOS-open drain output. **[6]**

b) Draw 2 i/p standard TTL NAND gate with Totem pole. Explain operation of transistor. (ON/OFF) with suitable input conditions and truth table. **[6]**

c) Explain the Tristate logic and Tristate TTL inverter. **[6]**

OR

Q6) a) Explain TTL open collector. **[6]**

b) Define the following terms and mention the standard values for TTL logic Family : **[6]**

i) Noise Margin

ii) Fan Out

iii) Power Dissipation

c) Draw and explain the circuit diagram of CMOS Inverter. **[6]**

Q7) a) Explain the Memory organization of the microprocessor. **[6]**

b) Draw and explain 4-bit Multiplier circuit using ALU and shift registers. **[6]**

c) Explain the basic arithmetic operations can be performed by using IC 74181. **[5]**

OR

Q8) a) What is Microprocessor? Explain the system bus in brief. **[6]**

b) Draw and explain the block diagram of Microprocessor with its Functional units. **[6]**

c) What are various functional units of microprocessors? Explain ALU in brief. **[5]**

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